03 · Domain 3 — Cloud Technology and Services (34%)

The **biggest domain**. It is split in three parts for readability.

> **Scope tags (official guide v1.3):** `v1.3` listed in the latest guide · `LEGACY` = older guide only / not listed now · `OUT` explicitly out of scope · `NEW` new in the v1.3 list. See file 00 for the full scope map.

Part	Content	Task statements
03a — Deploy, global infrastructure, compute	Ways to access AWS, IaC, deployment models, connectivity, Regions/AZs/edge, EC2, containers, serverless, scaling, load balancing	3.1 · 3.2 · 3.3
03b — Databases, networking, storage	RDS/Aurora/DynamoDB..., VPC/Route 53/CloudFront..., S3/EBS/EFS/FSx...	3.4 · 3.5 · 3.6
03c — AI/ML, analytics, integration and other services	SageMaker AI, Lex, Athena, Kinesis, SNS/SQS, dev tools, management & governance, IoT, end-user computing	3.7 · 3.8

The mental map of AWS

The "which service?" master table

If the question says...	Answer
Virtual servers in the cloud	**EC2**
Run code without managing servers, event-driven	**Lambda**
Containers without managing servers	**Fargate**
Container orchestration (AWS-native) / Kubernetes	**ECS** / **EKS**
Deploy a web app, AWS handles infra (upload code)	**Elastic Beanstalk**
Simple, low-cost VPS for small apps/blogs	**Lightsail**
Distribute incoming traffic across targets	**Elastic Load Balancing**
Add/remove instances automatically	**EC2 Auto Scaling**
Managed relational database	**RDS** (or **Aurora** for high performance)
Serverless NoSQL key-value, single-digit ms	**DynamoDB**
In-memory cache / microsecond reads	**ElastiCache** / **MemoryDB**
Data warehouse / analytics on petabytes	**Redshift**
Graph database	**Neptune**
Migrate databases	**DMS** (+ **SCT** to convert schema)
Object storage (files, backups, static site, data lake)	**S3**
Block storage for an EC2 instance	**EBS**
Shared file system for Linux (NFS)	**EFS**
Windows file shares / high-performance Lustre	**FSx**
On-prem to cloud storage gateway	**Storage Gateway**
Centralized backup	**AWS Backup**
Private isolated network	**VPC**
DNS	**Route 53**
CDN / cache at edge	**CloudFront**
Static anycast IPs, AWS backbone routing	**Global Accelerator**
Dedicated private connection to AWS	**Direct Connect**
Encrypted tunnel over internet	**AWS VPN**
Create/publish APIs	**API Gateway**
Decouple (queue)	**SQS**
Pub/sub notifications (email, SMS, push)	**SNS**
Event bus / schedule / SaaS events	**EventBridge**
Orchestrate workflows	**Step Functions**
Build/train/deploy ML models	**SageMaker AI** (name in v1.3; formerly SageMaker)
Chatbot	**Lex**
Enterprise search	**Kendra** `LEGACY` (not on v1.3 list)
Images/video analysis	**Rekognition**
Text to speech / speech to text	**Polly** / **Transcribe**

If the question says…	Answer
Translate text	**Translate**
Extract text from scanned documents	**Textract**
Sentiment / entities in text	**Comprehend**
Query S3 with SQL	**Athena**
Real-time streaming data	**Kinesis**
ETL / data catalog	**Glue**
BI dashboards	**Amazon Quick** (v1.3 name; formerly QuickSight)
Infrastructure as code	**CloudFormation**
Monitor metrics/logs/alarms	**CloudWatch**
Audit API activity	**CloudTrail**
Manage many accounts	**Organizations** (+ **Control Tower**)
Virtual desktops	**WorkSpaces**
Streaming desktop apps	**AppStream 2.0**
IoT device connectivity	**IoT Core**
Cloud contact center	**Connect**
Send bulk email	**SES**

03a · Deploying, Global Infrastructure and Compute

Task statements: **3.1** Methods of deploying and operating · **3.2** AWS global infrastructure · **3.3** Compute services

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3.1 Methods of deploying and operating in the AWS Cloud

Ways to access and provision AWS services

Method	What it is	Use when
AWS Management Console	Web interface	Learning, one-time/manual tasks, exploring
AWS CLI	Command-line tool (also **AWS CloudShell** — browser-based shell; `OUT` out of scope in v1.3)	Scripting, automation, repeatable tasks
AWS SDKs	Libraries for programming languages	Applications calling AWS services programmatically
AWS APIs	Underlying HTTP APIs (everything above uses them)	Direct programmatic access
Infrastructure as Code (IaC)	Describe infrastructure in templates	**Repeatable**, consistent, versioned deployments

IaC services

- **AWS CloudFormation**: define resources in **JSON/YAML templates** → creates a **stack**; repeat across Regions/accounts; rollback on failure; free (you pay for the resources).
- **AWS CDK (Cloud Development Kit)**: define infra using programming languages; generates CloudFormation.
- **AWS Elastic Beanstalk**: upload code; AWS provisions and manages the environment.

One-time vs repeatable: console = one-time/manual; **IaC/CLI/SDK = repeatable processes** (fewer errors, scalable, auditable).

Deployment models (recap)

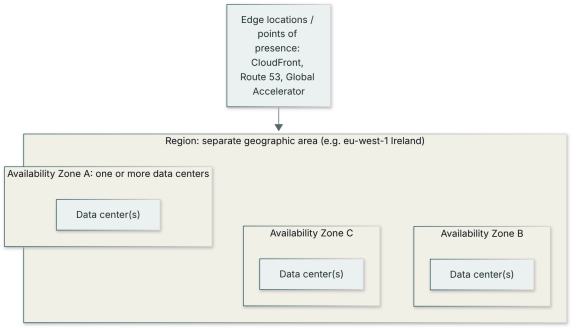
Cloud · **Hybrid** · **On-premises**. Hybrid enablers: Direct Connect, VPN, Storage Gateway, **AWS Outposts** (AWS infrastructure on-prem).

Connectivity options

Option	Description	Pros	Cons
Public internet	Access AWS public endpoints over HTTPS	Simple	Variable latency, no dedicated bandwidth
AWS VPN (Site-to-Site / Client VPN)	**Encrypted** IPsec tunnel **over the internet**	Fast to set up, low cost, encrypted	Variable performance (internet-dependent)
AWS Direct Connect	**Dedicated private network connection** from your site to AWS (via a Direct Connect location)	**Consistent** performance, higher bandwidth, lower data-transfer cost, private	Takes weeks to provision, higher cost; **not encrypted by default** (can combine with VPN)

> "Consistent network performance / dedicated private" → **Direct Connect**. "Quick, encrypted, over internet" → **VPN**.

3.2 AWS Global Infrastructure



Component	Definition	Key facts
Region	A **geographic area** containing **multiple isolated Availability Zones**	Each Region is **independent**; resources are **not automatically replicated** across Regions; **data stays in the Region** unless you move it
Availability Zone (AZ)	**One or more discrete data centers** with redundant power, networking, connectivity, in a Region	AZs are physically separated (but close, low-latency links); **do not share a single point of failure**; design for HA by using **multiple AZs**
Edge location (point of presence)	Sites used by **CloudFront** (cache content), **Route 53** (DNS), **Global Accelerator**, WAF/Shield at the edge	Many more edge locations than Regions; bring content **closer to users**
Regional edge cache	Larger cache between edge locations and origin	Keeps content longer
AWS Local Zones	Extension of a Region placing compute/storage **near large population/industry centers** for **single-digit-ms latency**	Example: media rendering, real-time gaming near a city LEGACY not on v1.3 list
AWS Wavelength Zones	AWS infrastructure embedded in **5G telecom networks**	Ultra-low latency for mobile apps OUT out of scope in v1.3
AWS Outposts	AWS-managed racks/servers **in your own data center**	Hybrid, local data processing, consistent AWS APIs v1.3

How to choose a Region (4 factors)

1. **Compliance / data governance** (legal requirements on where data resides)
2. **Latency** (closest to users)
3. **Service availability** (not all services/features in every Region)
4. **Pricing** (varies by Region)

Achieving high availability

- Use **multiple AZs**: e.g., EC2 instances in 2+ AZs behind a load balancer; **RDS Multi-AZ**.
- Use **multiple Regions** for: **disaster recovery**, **business continuity**, **low latency for global users**, **data sovereignty** needs, resilience to a Region-wide failure.
- AZ failure ≠ Region failure; design accordingly.

> Trap: **A Region has multiple AZs; an AZ has one or more data centers; edge locations are separate from AZs. CloudFront uses edge locations, not AZs.**

3.3 Compute services

Amazon EC2 (Elastic Compute Cloud) — virtual servers (IaaS)

Key concepts

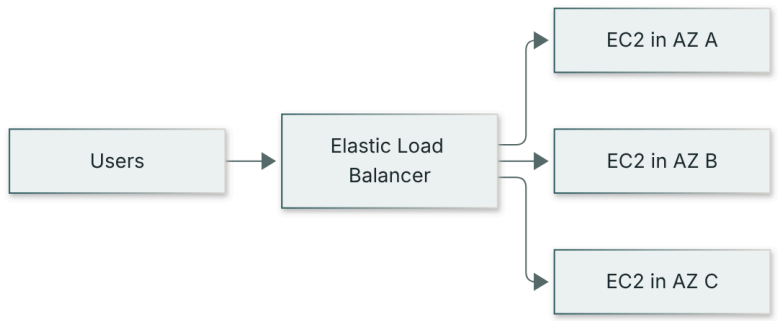
- **Instance** = virtual server · **AMI** (Amazon Machine Image) = template (OS + software) used to launch instances
- **Instance type** = combination of vCPU, memory, storage, network (e.g., `t3.micro`, `m5.large`, `c6g.xlarge`)
- **Key pair** = SSH/RDP login credentials · **Security group** = virtual firewall · **EBS** = persistent volumes · **User data** = bootstrap script at launch
- You pay per second/hour (Linux) while running; choose **purchasing option** (On-Demand, Reserved, Savings Plans, Spot… see Domain 4)
- You manage: **OS, patches, apps, security groups** (shared responsibility)
- **Regional** service; instance lives in **one AZ**

Instance families (what the letter means)

Family	Letters	Optimized for	Example use cases
General purpose	**M**, **T** (burstable)	Balance of compute/memory/network	Web servers, small DBs, dev/test
Compute optimized	**C**	High CPU performance	Batch processing, HPC, gaming servers, scientific modeling, media encoding
Memory optimized	**R**, **X**, **z**	Large memory	In-memory DBs, big data caches, real-time analytics
Storage optimized	**I**, **D**, **H**	High sequential/random I/O on local storage	NoSQL DBs, data warehousing, distributed file systems
Accelerated computing	**P**, **G**, **Inf**, **Trn**, **F**	GPUs / custom accelerators	ML training, graphics rendering, video transcoding

> **"C = Compute, R = RAM, I = I/O, G/P = GPU, T = Tiny/burstable, M = Medium/balanced."**

Elastic Load Balancing (ELB) — distribute traffic

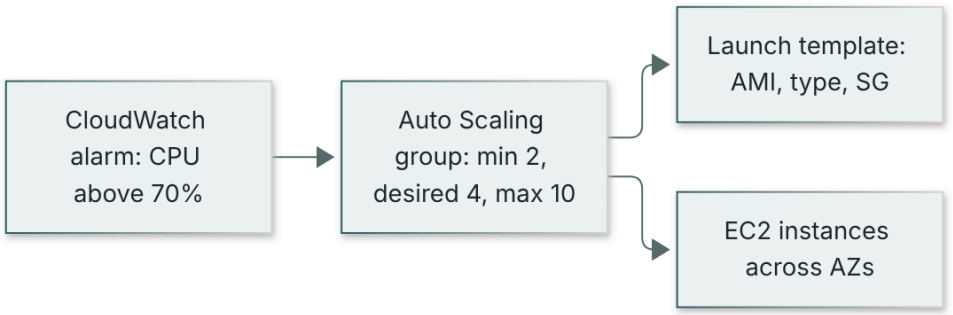


Type	Layer	Use
Application Load Balancer (ALB)	**Layer 7** (HTTP/HTTPS)	Path/host-based routing, microservices, containers
Network Load Balancer (NLB)	**Layer 4** (TCP/UDP)	Extreme performance, static IP, millions of requests/s
Gateway Load Balancer (GWLB)	Layer 3	Deploy/scale 3rd-party virtual appliances (firewalls)
(Classic Load Balancer)	L4/L7	Legacy

Load balancers perform **health checks** and send traffic only to healthy targets; spread across **AZs** → HA.

EC2 Auto Scaling — elasticity

Automatically **adds/removes EC2 instances** based on rules → **elasticity + cost savings + availability**.



Scaling types: **dynamic** (target tracking, step), **scheduled** (known traffic), **predictive** (ML-based), plus **manual/fixed** and **health-check replacement** (maintains desired capacity). Defined by **min / desired / max**. Auto Scaling itself is **free** (pay for the resources).

Containers

Service	What it does
Docker container	Package of app + dependencies; portable
Amazon ECS (Elastic Container Service)	**AWS-native** orchestration to run/manage Docker containers (on EC2 or Fargate)
Amazon EKS (Elastic Kubernetes Service)	**Managed Kubernetes** control plane; use when you want Kubernetes/portability
Amazon ECR (Elastic Container Registry)	**Registry** to store/manage container images
AWS Fargate	**Serverless compute for containers**: no EC2 to provision/manage; works with ECS and EKS; pay per vCPU/memory used

> ECS vs EKS: AWS proprietary simple vs open-source Kubernetes. Fargate vs EC2 launch type: serverless vs you manage the instances.

Serverless

AWS Lambda — run code **without provisioning/managing servers**.

- **Event-driven**: triggered by S3 uploads, API Gateway, DynamoDB, SQS, EventBridge schedules, etc.
- Pay only for **number of requests + duration (ms) — no charge when idle**
- **Automatic scaling**; max run time **15 minutes** per invocation; choose memory (CPU scales with it)
- You manage: code + IAM; AWS manages servers/OS/runtime
- Good for: small, short tasks, image processing, backends, automation

Compute service	When to choose
EC2	Full control of OS, long-running/stateful workloads, custom software
Lambda	Short event-driven functions, no servers
Fargate	Containers, no server management
ECS / EKS (EC2)	Containers with control over hosts
Elastic Beanstalk	Upload code (Java, .NET, Python, Node...) → AWS deploys with EC2, ELB, Auto Scaling for you (PaaS-like)
Lightsail	Simple, **fixed low monthly price** VPS, WordPress sites, small apps; for beginners
AWS Batch	Run **batch jobs** at any scale; provisions optimal compute automatically
AWS Outposts	AWS services on-premises
AWS Local Zones / Wavelength	Low-latency edge compute

Compute decision flow

Self-check

1. Console vs CLI vs SDK vs CloudFormation — when each?
2. VPN vs Direct Connect?
3. Region vs AZ vs edge location vs Local Zone vs Outposts?
4. Four factors to choose a Region?
5. Instance family for: in-memory DB? ML training? batch processing? NoSQL with high I/O?
6. ALB vs NLB layer? Which service gives elasticity?
7. ECS vs EKS vs Fargate vs ECR?
8. Max Lambda duration? Does Lambda cost anything when idle?

Answers

1. Console = manual; CLI = scripts; SDK = code; CloudFormation = repeatable IaC.
2. VPN = encrypted over internet, quick; Direct Connect = dedicated private, consistent.
3. See table.
4. Compliance, latency, service availability, price.
5. Memory optimized (R/X) · Accelerated (P/G) · Compute optimized (C) · Storage optimized (I).
6. ALB = L7, NLB = L4 · Auto Scaling.
7. ECS = AWS-native orchestration · EKS = Kubernetes · Fargate = serverless compute for containers · ECR = image registry.
8. 15 min · No (pay per request/duration).

03b · Databases, Networking and Storage

Task statements: **3.4** Database services · **3.5** Network services · **3.6** Storage services

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3.4 Database services

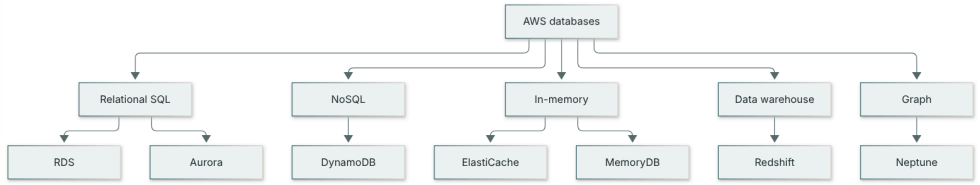
Decision: self-managed on EC2 vs AWS managed database

	DB on EC2 (self-managed)	Managed DB (RDS, DynamoDB...)
OS & DB patching	You	**AWS**

	DB on EC2 (self-managed)	Managed DB (RDS, DynamoDB…)
Backups, HA, failover	You build it	**Built in** (Multi-AZ, automated backups)
Scaling	Manual	Easier / automatic
Control	Full (OS access, any version/engine)	Limited (no OS access)
Effort/cost of ops	High	**Low**

→ Choose EC2-hosted only if you need OS-level control or an unsupported engine. Otherwise **managed**.

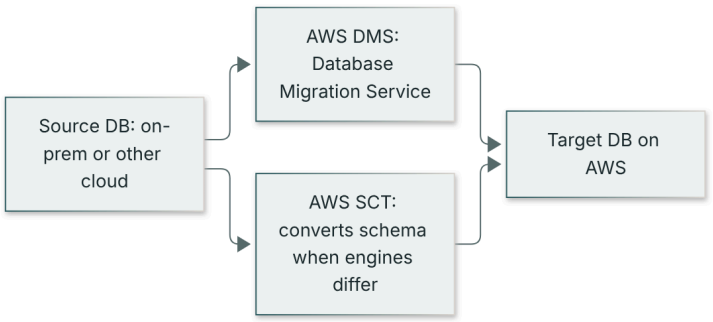
Database types and services



Service	Type	Key facts
Amazon RDS	**Managed relational**	Engines: **MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Db2**(+ Aurora). AWS handles patching, backups, **Multi-AZ** (HA, automatic failover), **Read Replicas** (read performance scaling). You can't SSH to the instance
Amazon Aurora	Relational, **MySQL- and PostgreSQL-compatible**, cloud-built	Up to ~**5× MySQL / 3× PostgreSQL** performance claims; storage auto-scales; **6 copies of data across 3 AZs**; ~1/10 cost of commercial DBs; **Aurora Serverless** auto-scales
Amazon DynamoDB	**NoSQL key-value & document**, **serverless**	**Single-digit millisecond** latency at any scale; fully managed; **Global Tables** (multi-Region); **on-demand** or provisioned capacity; **DAX** = in-memory cache for DynamoDB
Amazon ElastiCache	**In-memory cache** (**Redis / Memcached**)	Sub-ms latency; reduce DB load; sessions, leaderboards
Amazon MemoryDB (for Redis)	**Durable in-memory database** (Redis-compatible)	Primary DB with microsecond reads, multi-AZ durability OUT out of scope in v1.3 (ElastiCache v1.3 is the in-scope memory DB)
Amazon Redshift	**Data warehouse** (columnar, OLAP)	Analytics over **petabytes**; SQL; BI; not for transactional OLTP
Amazon Neptune	**Graph database**	Relationships: social networks, fraud detection, knowledge graphs
Amazon DocumentDB	Document DB, **MongoDB-compatible**	JSON-like documents v1.3 on v1.3 list
Amazon QLDB (ledger)	Immutable ledger	(Check current availability) (beyond scope)
Amazon Keyspaces	Cassandra-compatible	OUT out of scope
Amazon Timestream	Time-series	IoT/metrics (beyond scope)

> Exam pairs: **relational + managed → RDS**, **relational + highest performance/cloud-native → Aurora**, **NoSQL + serverless + ms → DynamoDB**, **cache → ElastiCache**, **warehouse/analytics → Redshift**, **graph → Neptune**.

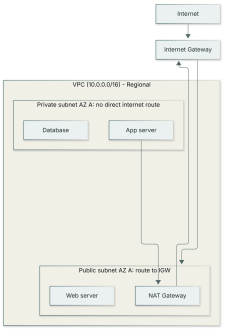
Database migration



- **AWS DMS**: migrates databases to AWS **with minimal downtime**; source stays operational; supports **homogeneous** (MySQL→MySQL) and **heterogeneous** (Oracle→Aurora) migrations; continuous replication.
- **AWS SCT**: **converts the schema and code** from one engine to another (used with DMS for heterogeneous migrations).

3.5 Network services

Amazon VPC (Virtual Private Cloud) — your private network in AWS



Component	Role
VPC	Logically isolated virtual network in a **Region**; you define the IP range (CIDR)
Subnet	A range of IPs **within one AZ**. **Public subnet**: route to an Internet Gateway. **Private subnet**: no direct route to the internet
Internet Gateway (IGW)	Connects the VPC to the internet (inbound/outbound)
NAT Gateway	Lets **private-subnet** resources **initiate outbound** internet traffic (e.g., updates) without being reachable from the internet
Route table	Rules that decide where network traffic goes
Security Group	Instance-level stateful firewall (allow only)
Network ACL	Subnet-level stateless firewall (allow + deny)
VPC Peering	Private connection between two VPCs
Transit Gateway	Hub to connect many VPCs and on-prem networks v1.3 NEW v1.3
VPC Endpoints / PrivateLink	Private access to AWS services **without traversing the internet** (Gateway endpoints for S3/DynamoDB; Interface endpoints) (PrivateLink v1.3 NEW v1.3)
VPN Gateway / Direct Connect Gateway	Connect to on-premises

A **default VPC** exists in each Region so you can launch EC2 immediately.

Other networking and content-delivery services

Service	What it does	Remember
Amazon Route 53	**Scalable DNS** + domain registration + **health checks**	Routing policies: **simple, weighted, latency-based, failover, geolocation**, geoproximity. The "53" is the DNS port
Amazon CloudFront	**CDN**: caches content at **edge locations** → lower latency for users	Origins: S3, ALB, EC2, custom; integrates with WAF/Shield; **HTTPS**
AWS Global Accelerator	Gives **2 static anycast IPs**, routes user traffic over the **AWS global network** to the best healthy endpoint	Improves availability/performance for **TCP/UDP** apps; not caching
Amazon API Gateway	Create, publish, secure, monitor **APIs** (REST, HTTP, WebSocket)	Common front for **Lambda**
AWS Direct Connect	Dedicated private connection	See 03a
AWS VPN	Encrypted tunnel	See 03a
AWS Transit Gateway	Network hub	v1.3 NEW v1.3
Elastic Load Balancing	Traffic distribution	See 03a

> CloudFront vs Global Accelerator: **CloudFront caches content (HTTP/S, static+dynamic)**; **Global Accelerator doesn't cache — it optimizes the network path** (static IPs, fast failover).

3.6 Storage services

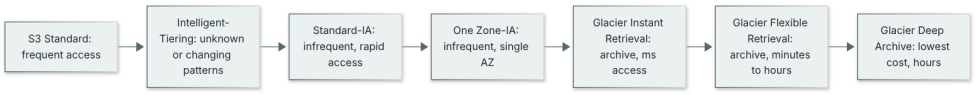
The three storage types

Type	Example	Access	Best for
Object	**S3**	HTTP API (key = object name)	Files, backups, media, data lakes, static sites — unlimited scale
Block	**EBS**, instance store	Attached to an EC2 as a **disk**	OS volumes, databases, low-latency
File	**EFS**, **FSx**	Network file system (NFS/SMB)	Shared files across many servers

Amazon S3 (Simple Storage Service)

- **Object storage** in **buckets** (globally unique names; bucket created in a Region). Object up to **5 TB**; single PUT up to 5 GB (multipart for larger).
- **Durability: 99.999999999% (11 nines)**; Standard availability 99.99% (design target).
- Features: **versioning**, **replication** (CRR/SRR), **encryption** (SSE-S3, SSE-KMS, SSE-C), **bucket policies / ACLs**, **Block Public Access**, **static website hosting**, **lifecycle policies**, **Object Lock** (WORM), **Transfer Acceleration**, event notifications, access logs.
- Pay for **storage + requests + data transfer out** (and retrieval for cold classes).

S3 storage classes — from hot to cold



Class	Use case	AZs	Min storage duration	Retrieval
S3 Standard	Frequently accessed, general	≥3	none	ms
S3 Intelligent-Tiering	Unknown/changing access; **auto-moves** objects between tiers	≥3	none	ms (+ small monitoring fee)
S3 Standard-IA	Infrequent access, need rapid access; lower storage cost, **retrieval fee**	≥3	**30 days**	ms
S3 One Zone-IA	Infrequent, **re-creatable** data; stored in **one AZ** (lost if AZ lost)	**1**	30 days	ms
S3 Glacier Instant Retrieval	Archive, rarely accessed but needs **millisecond** access	≥3	90 days	ms
S3 Glacier Flexible Retrieval (formerly Glacier)	Archive; retrieval **minutes to hours**	≥3	90 days	min–hours
S3 Glacier Deep Archive	**Cheapest**; long-term retention (compliance, 7–10+ years)	≥3	**180 days**	hours (~12–48 h)

(There are also S3 Express One Zone for ultra-low latency — beyond the exam's core.)

S3 Lifecycle policies: rules that **automatically transition** objects between classes (Standard → IA → Glacier) after N days and/or **expire (delete)** them. Use to cut cost automatically.

Amazon EBS (Elastic Block Store)

- **Persistent block storage** for EC2; **lives independent of the instance** (data survives stop/termination if configured); **attached to one instance in the same AZ** (Multi-Attach for some types).
- **Snapshots**: incremental backups stored in S3; copy across AZ/Region; restore to new volumes.
- Volume types: **SSD** (gp3/gp2 general purpose, io2/io1 Provisioned IOPS for DBs) and **HDD** (st1 throughput-optimized, sc1 cold).
- Encryption with KMS.

EC2 Instance Store

Temporary block storage physically attached to the host; **very high performance** but **data is lost** when the instance stops, terminates, or hardware fails → for caches, buffers, scratch data.

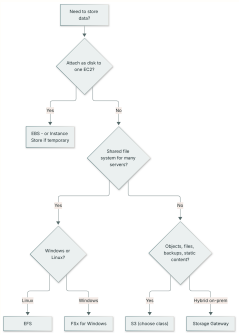
File storage

Service	What	Use
Amazon EFS	**Managed NFS** file system for **Linux**; **elastic** (grows/shrinks automatically); **multi-AZ**; shared by **thousands of EC2/containers/Lambda**	Content management, shared home dirs, web serving
Amazon FSx	Managed 3rd-party file systems: **FSx for Windows File Server** (SMB, Active Directory), **FSx for Lustre** (HPC/ML; OUT out of scope in v1.3), **FSx for NetApp ONTAP**, **FSx for OpenZFS**	Windows shares; high-performance computing

Hybrid / transfer / backup / DR

Service	Purpose
AWS Storage Gateway	**Hybrid storage** — on-prem apps use cloud storage with **local cache** (File, Volume, Tape gateway)
AWS Backup	**Centralized, policy-based backup** across services (EBS, RDS, DynamoDB, EFS, S3, etc.), cross-Region/account, compliance reporting
AWS Elastic Disaster Recovery (DRS)	Fast recovery of on-prem/cloud servers into AWS
AWS Snow Family	**Physical devices** to move data offline / edge compute LEGACY not on v1.3 list
AWS DataSync	Automated online transfer on-prem AWS storage LEGACY not on v1.3 list
AWS Transfer Family	SFTP/FTPS/FTP into S3/EFS OUT out of scope in v1.3

Storage decision flow



Self-check

1. RDS vs Aurora vs DynamoDB vs Redshift vs ElastiCache vs Neptune — one-line each.
2. DMS vs SCT?
3. Public vs private subnet? What does a NAT Gateway do? IGW?
4. CloudFront vs Global Accelerator vs Route 53?
5. S3 classes: which is cheapest? which needs ms retrieval but rarely accessed? which auto-moves objects? which is single-AZ?
6. EBS vs Instance Store vs EFS vs S3?
7. What do lifecycle policies do? What is S3 durability?
8. Which service centrally backs up many AWS services?

Answers

1. RDS managed relational · Aurora cloud-native MySQL/Postgres-compatible · DynamoDB serverless NoSQL · Redshift data warehouse · ElastiCache in-memory cache · Neptune graph.
2. DMS migrates data (minimal downtime); SCT converts schema.
3. Public = route to IGW; private = no direct internet; NAT = outbound only for private subnets; IGW = VPCinternet.
4. CloudFront = CDN cache; Global Accelerator = static IPs/AWS backbone; Route 53 = DNS.
5. Deep Archive · Glacier Instant Retrieval · Intelligent-Tiering · One Zone-IA.
6. EBS = persistent block for EC2; Instance Store = ephemeral block; EFS = shared Linux files; S3 = objects.
7. Auto transition/expire objects; 11 nines.
8. AWS Backup.

03c · AI/ML, Analytics and Other In-Scope Services

Task statements: **3.7** AI/ML and analytics services · **3.8** Other in-scope service categories

> **Scope tags (official guide v1.3):** `v1.3` listed in the latest guide · `LEGACY` = older guide only / not listed now · `OUT` explicitly out of scope · `NEW` new in the v1.3 list. See file 00 for the full scope map.

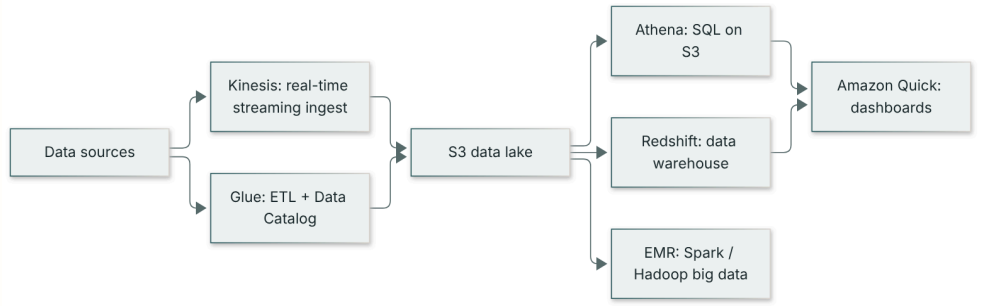
3.7 AI/ML services

Service	What it does	Typical exam phrase
Amazon SageMaker AI (v1.3 name; formerly SageMaker) `v1.3`	Fully managed platform to **build, train and deploy ML models** (notebooks, data labeling, training, hosting)	"build/train/deploy custom machine learning models"
Amazon Rekognition	**Image and video analysis**: object/face detection, facial comparison, content moderation	"identify objects/faces in images"
Amazon Textract	**Extracts text, forms and tables** from scanned documents (beyond simple OCR)	"extract data from scanned invoices/forms"
Amazon Comprehend	**NLP**: sentiment, entities, key phrases, language detection	"analyze sentiment of customer reviews"
Amazon Translate	**Language translation** (neural machine translation)	"translate text"
Amazon Polly	**Text → lifelike speech**	"convert text to speech"
Amazon Transcribe	**Speech → text** (automatic speech recognition)	"transcribe call recordings"
Amazon Lex	Build **conversational interfaces / chatbots** (voice & text; same tech as Alexa)	"build a chatbot"

Service	What it does	Typical exam phrase
Amazon Kendra	**Intelligent enterprise search** powered by ML (natural-language questions over your documents)	"search across company documents" LEGACY — not on v1.3 list
Amazon Bedrock	Managed access to **foundation models (generative AI)** via API	"build generative AI apps with FMs" *(newer service — may be beyond the guide's list but appears in modern AWS content)* (not on the v1.3 list — know only that it exists)
Amazon Q	Generative-AI assistant for business/developers	(modern context) v1.3 on v1.3 list

> Trick: **Polly = parrot (speaks). Transcribe = transcript (writes). Rekognition = recognition (sees). Lex = Alexa. Comprehend = comprehends text. Textract = extracts from documents.**

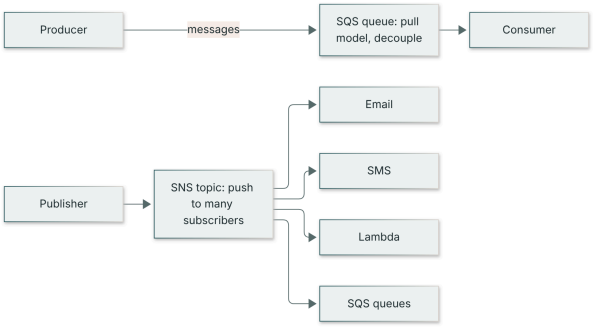
3.7 Analytics services



Service	What it does	Remember
Amazon Athena	**Serverless interactive SQL** queries **directly on data in S3**	Pay **per query (data scanned)**; no infrastructure
Amazon Kinesis	**Real-time streaming data** ingestion and processing (Data Streams, Data Firehose, Managed Flink, Video Streams)	Logs, clickstreams, IoT telemetry in real time
AWS Glue	Serverless **ETL** (extract-transform-load) + **Data Catalog**	Prepare/discover data
Amazon Quick (v1.3 name; formerly Amazon QuickSight) v1.3	**Business intelligence**: dashboards & visualizations, ML insights, pay-per-session	"create dashboards for business users"
Amazon EMR	Managed **big data** frameworks (**Hadoop, Spark**, Hive, Presto)	Process vast datasets
Amazon Redshift	Data warehouse	See 03b
Amazon OpenSearch Service	Search, log analytics, observability (Elasticsearch-derived)	Log search and analysis
Amazon MSK	Managed **Apache Kafka** streaming	Existing Kafka workloads OUT out of scope in v1.3
AWS Data Exchange	Find/subscribe/use **third-party data sets**	Buy external data OUT out of scope in v1.3
AWS Lake Formation	Build and secure data lakes	(not on list)

3.8 Other in-scope service categories

Application integration — decouple and connect



Service	Role	Key facts
Amazon SQS (Simple Queue Service)	**Message queue** to **decouple** components	**Pull** model (consumers poll); messages persist until processed; Standard (at-least-once, best-effort ordering) / FIFO (ordering) queues
Amazon SNS (Simple Notification Service)	**Pub/sub** messaging and **notifications**	**Push** to many subscribers: email, SMS, mobile push, HTTP, Lambda, SQS
Amazon EventBridge	**Serverless event bus**: route events from AWS services, your apps, and **SaaS partners**; **scheduled rules** (cron)	Successor of CloudWatch Events

Service	Role	Key facts
AWS Step Functions	**Serverless workflow orchestration** (state machines) coordinating Lambda & services	Visual workflows
Amazon MQ	Managed ActiveMQ/RabbitMQ	Migrate existing brokers

> **SQS = queue (pull, decouple). SNS = broadcast (push).** Classic combo: **SNS → multiple SQS (fan-out)**.

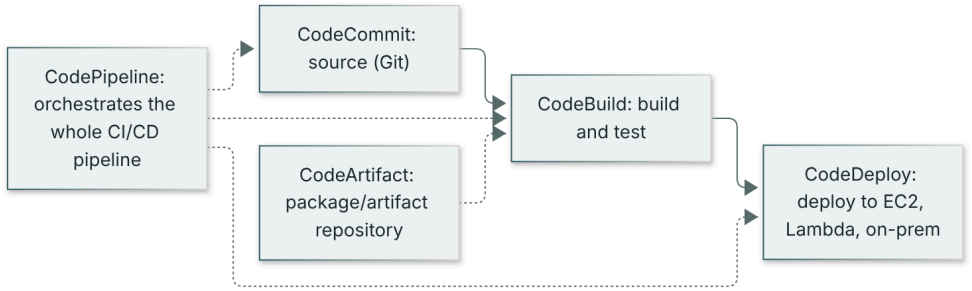
Business applications

- **Amazon Connect** — **cloud contact center** (omnichannel customer service, pay-as-you-go)
- **Amazon SES** (Simple Email Service) — send **bulk/transactional/marketing email** at scale

Customer engagement and support (also Domain 4)

- **AWS Activate for Startups** OUT (out of scope in v1.3) — credits, training, support for startups
- **AWS IQ** OUT (out of scope in v1.3) — hire **AWS Certified third-party experts** for projects
- **AWS Managed Services (AMS)** OUT (out of scope in v1.3) — AWS operates your infrastructure day-to-day
- **AWS Support** — Basic, Developer, Business, Enterprise On-Ramp, Enterprise

Developer tools



Service	Purpose
AWS CodeCommit	Managed private **Git repositories** *(AWS stopped onboarding new customers in 2024; know it for the exam)* LEGACY not on v1.3 list
AWS CodeBuild	**Compile, test, package** code (serverless build) v1.3
AWS CodeDeploy	**Automate deployments** to EC2, Lambda, ECS, on-prem OUT out of scope in v1.3
AWS CodePipeline	**CI/CD orchestration** (release pipeline) v1.3
AWS CodeArtifact	Secure **artifact/package repository** (npm, Maven, pip...) OUT out of scope in v1.3
AWS CodeStar	Unified project setup/management of dev toolchain *(retired for new use; know for the exam)* LEGACY not on v1.3 list
AWS Cloud9	Cloud-based **IDE** *(legacy for new customers)* LEGACY not on v1.3 list
AWS CloudShell	Browser-based shell with CLI pre-authenticated OUT out of scope in v1.3
AWS AppConfig	Deploy **application configuration/feature flags** safely OUT out of scope in v1.3
AWS X-Ray	**Distributed tracing**: analyze/debug requests across microservices (find bottlenecks) v1.3

End-user computing

Service	Purpose
Amazon WorkSpaces	**Virtual desktops (DaaS)** — Windows/Linux in the cloud
Amazon AppStream 2.0	**Stream desktop applications** to a browser (no full desktop)
Amazon WorkSpaces Secure Browser (v1.3 name; formerly WorkSpaces Web) v1.3	Secure **browser access** to internal websites/SaaS

→ These services **present the output of VMs/applications on end-user machines**.

Frontend web and mobile

- **AWS Amplify** v1.3 — build, deploy and host **full-stack web/mobile apps** quickly
- **AWS AppSync** — managed **GraphQL** API (real-time/offline data) LEGACY not on v1.3 list
- **AWS Device Farm** — test apps on real mobile devices/browsers OUT out of scope in v1.3

Internet of Things (IoT)

- **AWS IoT Core** v1.3 — connect **IoT devices** to the cloud securely (MQTT), route messages
- **AWS IoT Greengrass** OUT (out of scope in v1.3) — run **IoT logic locally at the edge** on devices, sync with cloud

Management and governance (very testable)

Service	One-line purpose
AWS Management Console	Web UI for managing AWS
AWS Organizations	**Centrally manage multiple accounts**: **consolidated billing**, **OUs**, **SCPs** (guardrails)
AWS Control Tower	Set up and govern a **multi-account landing zone** with best-practice guardrails
AWS CloudFormation	Infrastructure as code (stacks)
AWS Systems Manager	**Operational hub**: patch management, run commands, Parameter Store, Session Manager across fleets
Amazon CloudWatch	Metrics, logs, alarms, dashboards
AWS CloudTrail	API audit log
AWS Config	Resource configuration history & compliance rules
Service Quotas v1.3 NEW	View and manage your **service quotas (limits)** across AWS services, request increases
AWS Service Catalog	Catalog of **approved IT products** users can self-provision (governance)
AWS License Manager	Track and manage **software licenses** (BYOL)
AWS Launch Wizard	Guided deployment of **enterprise apps** (SAP, SQL Server) OUT out of scope in v1.3
AWS Compute Optimizer	ML-based **rightsizing recommendations**
AWS Resource Groups & Tag Editor	Group resources by tags; edit tags in bulk LEGACY not on v1.3 list
AWS Health Dashboard	**Service health + personalized events affecting your resources**
AWS Trusted Advisor	Best-practice checks (cost, performance, security, fault tolerance, limits)
AWS Well-Architected Tool	Review workloads against the 6 pillars
AWS Auto Scaling	Scaling for multiple resources (EC2, DynamoDB, Aurora...)

Disambiguation cheat — services that sound alike

Pair	Difference
SQS vs SNS	Queue (pull, decouple) vs pub/sub (push to many)
SNS vs SES	General notifications (SMS/push/email) vs bulk email marketing/transactional
EventBridge vs SNS	Event routing/rules/schedules & SaaS events vs simple fan-out notifications
Athena vs Redshift	Serverless SQL on S3 (ad hoc) vs data warehouse (repeated heavy analytics)
Kinesis vs SQS	Real-time streaming of ordered, replayable data vs message queue
Glue vs EMR	Serverless ETL/catalog vs big-data cluster frameworks
Amazon Quick (ex-QuickSight) vs Athena	Dashboards vs queries
Lex vs Polly vs Transcribe	Chatbot vs text→speech vs speech→text
Rekognition vs Textract	Objects/faces in images vs text/forms in documents
Kendra vs OpenSearch	LEGACY Kendra (legacy, not on v1.3 list): ML natural-language enterprise search · OpenSearch v1.3: search/log analytics engine
WorkSpaces vs AppStream 2.0 vs WorkSpaces Secure Browser	Whole virtual desktop · streaming individual apps · secure browser access
Amplify vs Elastic Beanstalk	Frontend/full-stack mobile & web hosting vs deploy backend app on managed infrastructure
Organizations vs Control Tower	Account management/billing/SCPs vs automated landing zone with guardrails
CloudWatch vs X-Ray	Metrics/logs/alarms vs request tracing
Systems Manager vs CloudFormation	Operate/patch/manage instances vs provision infrastructure

Self-check

1. Which AI service: chatbot · text-to-speech · speech-to-text · image analysis · sentiment · translate · document forms extraction · enterprise search · build custom models?
2. Athena vs Redshift vs Glue vs Amazon Quick vs Kinesis — one line each.
3. SQS vs SNS vs EventBridge vs Step Functions?
4. Which service lets customers call a contact center in the cloud?
5. Which tool traces requests across microservices?
6. WorkSpaces vs AppStream 2.0?
7. Organizations vs Control Tower vs Service Catalog?

Answers

1. Lex · Polly · Transcribe · Rekognition · Comprehend · Translate · Textract · Kendra (LEGACY legacy) · SageMaker AI.
2. SQL on S3 · warehouse · ETL/catalog · dashboards · real-time streaming.
3. Queue · pub/sub · event bus/schedules · workflow orchestration.
4. Amazon Connect.

5. AWS X-Ray.
6. Full desktop vs individual apps.
7. Multi-account mgmt & SCPs · landing zone guardrails · approved product catalog.